

200W Planar Transformer for Dual Active Bridge (DAB)

Electrical Specifications

- Input: 48 V
- Output: 107 V
- Power: 200 W
- Switching Frequency: 50 kHz

Transformer Design Parameters

- $L_m = 448.4 \mu H$
- $L_{lk} = 27.2 \mu H$
- Coupling $k \approx 0.993$
- Turns Ratio = 2.24
- Core: N87 Ferrite

Design Objectives

- Engineered leakage for power transfer
- High magnetic coupling
- Low core and copper losses
- Multiphysics validation

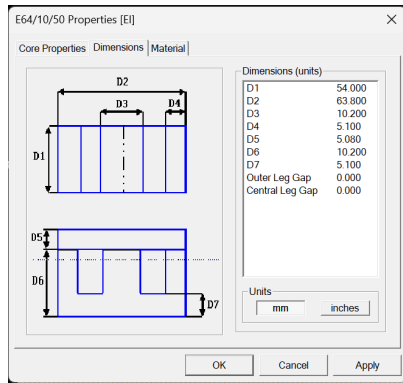
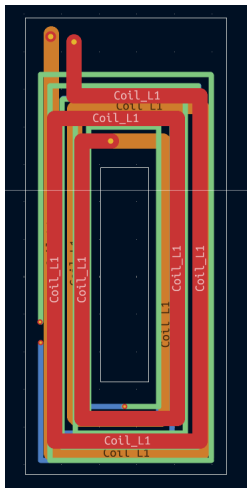
Area Product Verification

$$A_c A_w \geq \frac{L_p I_{peak} I_{RMS}}{B J K_w}$$

- Required: 117600 mm^4
- Core: 300685 mm^4

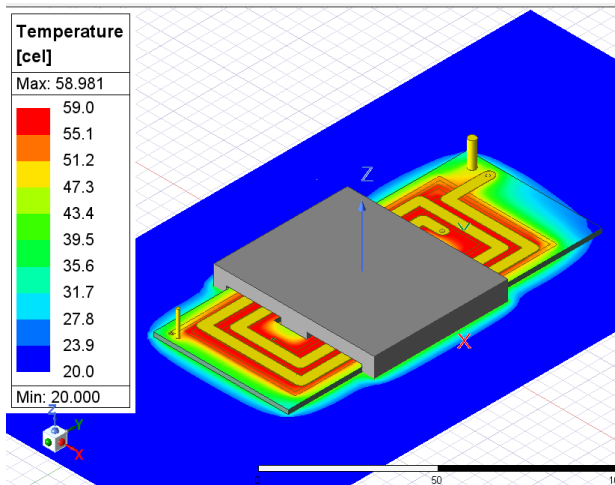
Design satisfies condition

Planar Transformer PCB & Core Geometry



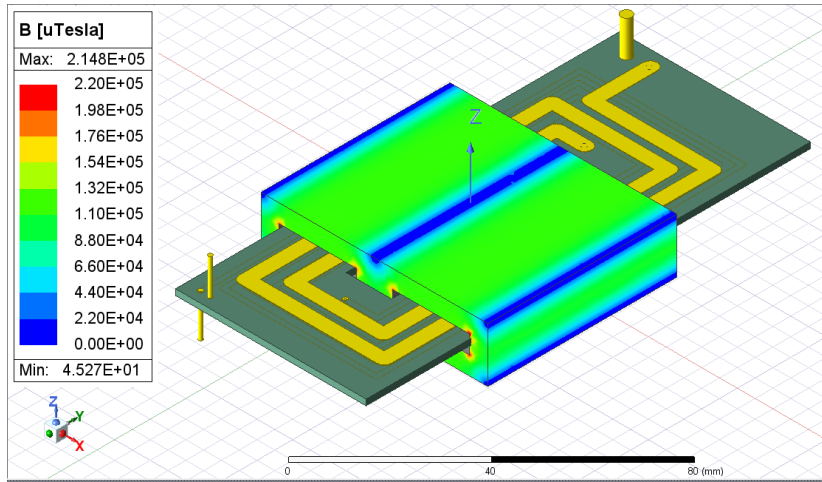
Core: E64/10/50 (EI)

Thermal Distribution (Icepak)



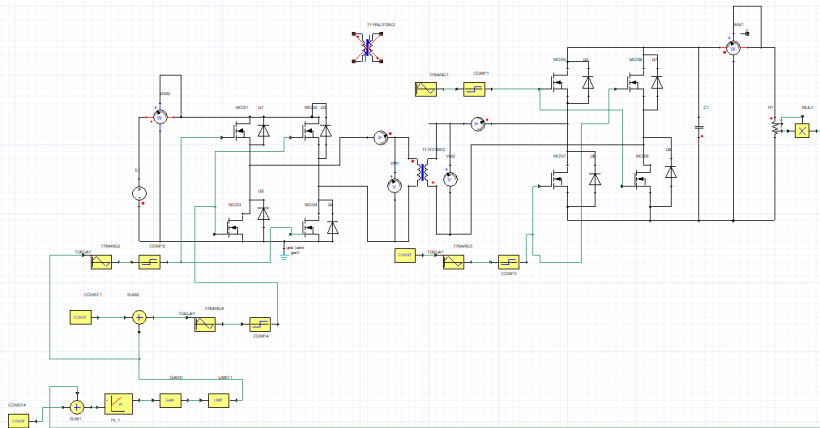
Max Temperature $\approx 59^{\circ}\text{C}$

Magnetic Flux Density (Maxwell 3D)



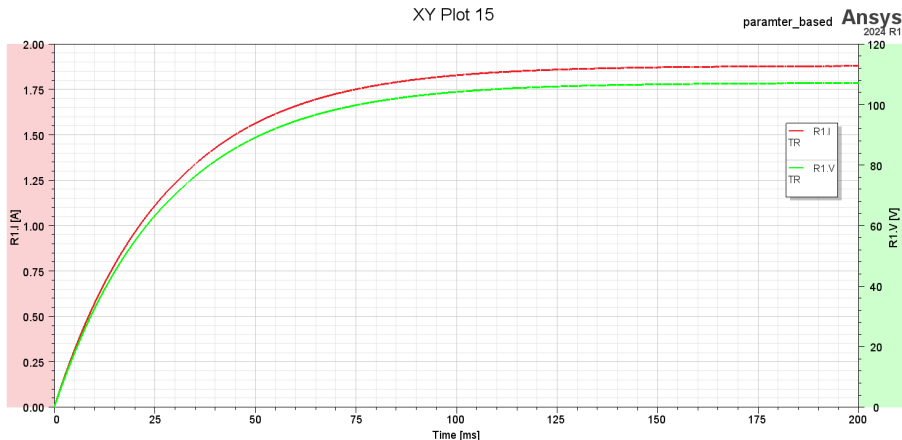
Peak Local Flux $\approx 0.21 \text{ T}$ (Localized)

DAB System Simulation (Simplorer1)



System-level validation of DAB converter operation

Output Validation



Validated 107V @ 1.87A (200W)

Key Results & Engineering Highlights

Performance

- Achieved 200W power transfer
- Transformer loss $\approx 3.6\text{ W}$
- Estimated Efficiency $\approx 98\%$
- Stable Phase-Shift Control

Engineering Strength

- Engineered Leakage Inductance
- High Magnetic Coupling ($k \approx 0.993$)
- Nonlinear Core Modeling
- Multiphysics Validation

Power Transfer in DAB:

$$P = \frac{N_p}{N_s} \frac{V_{Batt} V_{dc} \delta(\pi - |\delta|)}{2\pi\omega L_r}$$

Designed and achieved $L_{lk} = 27.2\text{ }\mu\text{H}$ for controlled 200W transfer at 50 kHz